

Supplementary Material: Incentive-Compatible Information Design*

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Abstract

This supplement contains omitted proofs and auxiliary results.

1 Proof of ??

Proof. The first and second forms below are the two allocations in ??. We use $m = \mathbf{E}H_0^p = p(1 - \ln p)$ as in the lemma. Recall from ?? that $\bar{q}(\mu) = \mu q_1(\mu) + (1 - \mu)q_0(\mu)$ is type μ 's probability of platform access; $\mathbf{E}[\bar{q}]$ averages this probability over types. The density is non-decreasing by ?? of the main paper. We first compare revenues across prices, then use the price and cutoff first-order conditions to bound α .

Allocations and envelope payments. For the first form,

$$(q_1, q_0) = \begin{cases} (0, 0), & \mu < \mu_0, \\ (\alpha, 1), & \mu_0 \leq \mu < \mu', \\ (1, 1), & \mu \geq \mu', \end{cases} \quad \alpha = \frac{p(1 - \mu_0)}{(1 - p)\mu_0}. \quad (1)$$

*The recent version of the paper can be found [here](#).

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The slope $K(\mu)$ is zero below μ_0 , p/μ_0 on $[\mu_0, \mu']$, and one above μ' . For $\mu \in [\mu_0, \mu']$, expected gross surplus after paying the seller is

$$(1-p)\alpha\mu - p(1-\mu) = \frac{p}{\mu_0}(\mu - \mu_0) = U(\mu). \quad (2)$$

The last equality follows from the envelope formula: exclusion below μ_0 and the normalization $U(0) = 0$ give $U(\mu_0) = 0$, while $K(\mu) = p/\mu_0$ on this interval. Gross surplus and information rents therefore coincide, so the fee is zero. The envelope formula above μ' instead gives fee $\mu'(1 - p/\mu_0)$.

Integrating the allocation in [Equation 1](#), using the mean constraint ?? of the main paper, and integrating these fees gives

$$\mathbf{E}[\bar{q}] = 1 - F(\mu_0) - (1-\alpha) \int_{\mu_0}^{\mu'} t dF(t), \quad (3)$$

$$(1-m)\mathbf{E}[\bar{q}] = \int_{\mu_0}^1 (1-t) dF(t), \quad (4)$$

$$\Pi = \left(1 - \frac{p}{\mu_0}\right) \mu' [1 - F(\mu')]. \quad (5)$$

For the second form in ??, $\mathbf{E}[\bar{q}] = \int_{\mu_0}^{\mu'} t dF(t) + 1 - F(\mu')$. Its fee is $(1-p)\mu_0$ for types in $[\mu_0, \mu']$ and $(1-p)\mu_0 - p(1-\mu')$ for types above μ' , so

$$\Pi = (1-p)\mu_0[1 - F(\mu_0)] - p(1-\mu')[1 - F(\mu')]. \quad (6)$$

Platform full participation in ?? of the main paper requires $\mu' \geq p$. Types in $[\mu_0, \mu']$ have posterior one in the second form and $\frac{\alpha\mu}{1-(1-\alpha)\mu}$ in the first; both have posterior μ above μ' .

A feasible second-form comparison. Consider the second form in ?? with $\mu_0 = \mu' = 3/5$: types below $3/5$ receive $(0, 0)$ and those above $3/5$ receive $(1, 1)$. We choose a price that makes this allocation feasible for the relaxed problem. Its revenue then provides a lower bound on optimal revenue, which we use to rule out the first form at low and high prices. The mean constraint for this allocation is

$$p'(1 - \ln p') = \frac{\int_{3/5}^1 t dF(t)}{1 - F(3/5)}. \quad (7)$$

The left side increases continuously from zero to one, so Equation 7 determines p' uniquely. For the uniform distribution, the right side is $4/5$. Since $\ln(25/11) > 9/11$, we have $p' < 11/25 < 3/5$. Thus platform full participation in ?? holds. The mean equality ?? and the crossing condition ?? also hold with $x = 0$, since the induced posterior law has no atom at p' . The envelope payments give fee $3/5 - p'$, and Equation 6 gives revenue exceeding

$$(3/5 - 11/25)(2/5) = 8/125 > 1/16. \quad (8)$$

Suppose $\|f - 1\|_\infty \leq \varepsilon \leq 1/100$. The density bounds give

$$1 - F\left(\frac{3}{5}\right) \geq \frac{2}{5}(1 - \varepsilon), \quad f(1) \leq 1 + \varepsilon. \quad (9)$$

The same bound controls the conditional mean in Equation 7. Since $\int_{3/5}^1 (t - 4/5) dt = 0$ and $\int_{3/5}^1 |t - 4/5| dt = 1/25$,

$$\begin{aligned} \frac{\int_{3/5}^1 t dF(t)}{1 - F(3/5)} - \frac{4}{5} &= \frac{\int_{3/5}^1 (t - 4/5)[f(t) - 1] dt}{1 - F(3/5)} \\ &\leq \frac{\varepsilon}{10(1 - \varepsilon)} \leq \frac{1}{990}. \end{aligned} \quad (10)$$

Thus the conditional mean is at most $793/990 < \frac{11}{25}[1 + \ln(25/11)]$. Because $p'(1 - \ln p')$ is increasing, Equation 7 implies $p' < 11/25 < 3/5$, ensuring platform full participation in ?. Using Equation 9, revenue satisfies

$$\begin{aligned} \left(\frac{3}{5} - p'\right) \left[1 - F\left(\frac{3}{5}\right)\right] &> \frac{8}{125}(1 - \varepsilon) \\ &> \frac{1 + \varepsilon}{16} \geq \frac{f(1)}{16}. \end{aligned} \quad (11)$$

The strict comparison follows from $\frac{8}{125}(1 - \varepsilon) - \frac{1 + \varepsilon}{16} = \frac{3 - 253\varepsilon}{2000} \geq \frac{47}{200000} > 0$.¹

Prices at most 3/10. We show that every positive-revenue mechanism of the first form earns less than $f(1)/16$ at these prices. By Equation 1,

$$\int_0^1 tq_1(t) dF(t) = \alpha \int_{\mu_0}^1 t dF(t) + (1 - \alpha) \int_{\mu'}^1 t dF(t). \quad (12)$$

¹This positive revenue and the bound $\Pi \leq -p \ln p$ keep maximizing sequences away from endpoint prices; compactness and continuity then give attainment.

Combining Equation 12 with the mean equality Equation 4, and using $\mu'[1 - F(\mu')] \leq \int_{\mu'}^1 t dF(t)$ in Equation 5 gives

$$\Pi \leq \frac{(1-p)m}{1-m} \int_{\mu_0}^1 (1-t) dF(t) - \frac{p(1-\mu_0)}{\mu_0} \int_{\mu_0}^1 t dF(t). \quad (13)$$

By ?? in the main appendix, f is non-decreasing. A type t above the midpoint $(1 + \mu_0)/2$ therefore has weakly greater density than $1 + \mu_0 - t$, which lies equally far below it. Their contributions to the mean relative to the midpoint sum to a nonnegative amount:

$$\begin{aligned} & \int_{\mu_0}^1 \left(t - \frac{1 + \mu_0}{2} \right) f(t) dt \\ &= \int_{(1+\mu_0)/2}^1 \left(t - \frac{1 + \mu_0}{2} \right) [f(t) - f(1 + \mu_0 - t)] dt \geq 0. \end{aligned} \quad (14)$$

Thus Equation 14 places the conditional mean above μ_0 at or above $(1 + \mu_0)/2$, equivalently

$$\int_{\mu_0}^1 t dF(t) \geq \frac{1 + \mu_0}{1 - \mu_0} \int_{\mu_0}^1 (1 - t) dF(t). \quad (15)$$

Substitute Equation 15 into Equation 13, and use $m - p = -p \ln p$. For positive revenue the resulting coefficient is positive, so the upper bound $\int_{\mu_0}^1 (1 - t) dF(t) \leq f(1)(1 - \mu_0)^2/2$ gives

$$\Pi \leq \frac{f(1)(1 - \mu_0)^2}{2} \left[\frac{-p \ln p}{1 - m} - \frac{p}{\mu_0} \right]. \quad (16)$$

For the low-price bound, use $\frac{dm}{dp} = -\ln p$ and the logarithm inequality $-\ln p \geq 2(1 - p)/(1 + p)$ in Equation 46. They imply

$$\frac{d}{dp} \left[\frac{-\sqrt{p} \ln p}{1 - m} \right] = \frac{p(\ln p)^2 - (1 + p) \ln p - 2(1 - p)}{2\sqrt{p}(1 - m)^2} > 0. \quad (17)$$

Evaluating the increasing ratio in Equation 17 at $p = 1/4$ gives, for $0 < p \leq 1/4$,

$$\frac{-p \ln p}{1 - m} \leq \frac{2 \ln 4}{3 - \ln 4} \sqrt{p} < \frac{16\sqrt{p}}{9}, \quad (18)$$

where the last inequality uses $\ln 4 < 24/17$. Applying Equation 18 to Equation 16 and completing the square gives

$$\frac{\Pi}{f(1)} \leq \frac{(1 - \mu_0)^2}{2} \left[\frac{16\sqrt{p}}{9} - \frac{p}{\mu_0} \right] \leq \frac{32\mu_0(1 - \mu_0)^2}{81} \leq \frac{128}{2187} < \frac{1}{16}. \quad (19)$$

The square completion bounds the bracket in Equation 19 by $64\mu_0/81$. The next bound uses only $0 < \mu_0 \leq 1$: the maximum of $\mu_0(1 - \mu_0)^2$ is $4/27$, attained at $\mu_0 = 1/3$.

For $1/4 \leq p \leq 3/10$, the ratio $(-\ln p)/(1 - m)$ is increasing as well. Here $m = p(1 - \ln p)$ varies with p ; including $dm/dp = -\ln p$ in the quotient rule gives

$$\frac{d}{dp} \frac{-\ln p}{1 - m} = \frac{(\ln p)^2 - (1 - m)/p}{(1 - m)^2} \geq 0. \quad (20)$$

Indeed, setting $t = -\ln p < 3/2$, the numerator is nonnegative by $e^t - 1 - t \leq t^2/[2(1 - t/3)] \leq t^2$. For each fixed μ_0 , the positive bracket in Equation 16 is therefore bounded by its value at $p = 3/10$. Since $\ln(10/3) < 35/29$, that price satisfies $(-p \ln p)/(1 - m) < 15/14$, and thus

$$\frac{\Pi}{f(1)} \leq \frac{(1 - \mu_0)^2}{2} \left[\frac{15}{14} - \frac{3}{10\mu_0} \right] < \frac{1}{16}. \quad (21)$$

To prove the strict bound in Equation 21, multiply the difference from $1/16$ by the positive denominator $560\mu_0$:

$$\begin{aligned} & 560\mu_0 \left\{ \frac{1}{16} - \frac{(1 - \mu_0)^2}{2} \left[\frac{15}{14} - \frac{3}{10\mu_0} \right] \right\} \\ &= (414 - 300\mu_0)(\mu_0 - 9/20)^2 + \frac{7}{20}\mu_0 + \frac{33}{200} > 0. \end{aligned} \quad (22)$$

Indeed, $0 < \mu_0 \leq 1$ implies $414 - 300\mu_0 \geq 114$; the square is nonnegative and the remaining terms are positive. Since $560\mu_0 > 0$, Equation 22 proves the desired inequality. Together, Equation 19 and Equation 21 exclude optimality of the first form at every $p \leq 3/10$.

Prices at least $1/2$. For $p \geq 1/2$, Equation 5 gives

$$\begin{aligned} \Pi &\leq (\mu' - p)[1 - F(\mu')] \leq f(1)(\mu' - p)(1 - \mu') \\ &\leq \frac{f(1)(1 - p)^2}{4} \leq \frac{f(1)}{16}. \end{aligned} \quad (23)$$

By Equation 11, the feasible second-form mechanism earns strictly more than either price range permits for the first form. An optimal relaxed mechanism of the first form must therefore have $3/10 < p < 1/2$, as asserted in ?? of the main paper.

Interior cutoffs and first-order conditions. Suppose an optimal relaxed mechanism has the first form. The preceding comparison then puts its price in $(3/10, 1/2)$. We use its constrained first-order conditions with respect to p and μ_0 to eliminate the multiplier and μ' , obtaining a relation between the price, the lower cutoff, and the density. This relation will imply $\mu_0 < 3p/(1 + 2p)$ and hence $\alpha > 1/3$, the bound used in ?? of the main appendix to establish the convex-order inequalities.

The comparison gives positive optimal revenue. If $\mu_0 = p$, the fee above μ' is also zero; if $\mu' = 1$, there is no positive mass of types paying that fee. Thus Equation 5 is zero in either boundary case. Since the lemma considers $\mu_0 < \mu'$, its cutoffs satisfy $0 < p < \mu_0 < \mu' < 1$ and $0 < \alpha < 1$.

Within these strict inequalities, Equation 1 ensures monotonicity and platform full participation in ?? of the main paper, with the latter binding at μ_0 . There is no atom at p , so ?? of the main paper holds with $x = 0$. Local variations in p, μ_0, μ' therefore need only preserve Equation 4. Substituting Equation 3 into that mean equality gives the Lagrangian, where λ is its multiplier:

$$\mathcal{L}(p, \mu_0, \mu'; \lambda) = \Pi - \lambda \left\{ (1 - m) \left[1 - F(\mu_0) - (1 - \alpha) \int_{\mu_0}^{\mu'} t dF(t) \right] - \int_{\mu_0}^1 (1 - t) dF(t) \right\}. \quad (24)$$

Holding p, μ_0 fixed, the derivative with respect to μ' of the mean-constraint expression in braces is

$$-(1 - m)(1 - \alpha)\mu' f(\mu') < 0. \quad (25)$$

By Equation 25, the mean equality locally determines μ' as a differentiable function of p and μ_0 . The feasible set therefore permits local variations in these two variables, and their Lagrange first-order conditions hold at the interior optimum.

To obtain these conditions, we differentiate Equation 24 with respect to p and μ_0 . In each partial derivative the other two choice variables and the prior are held

fixed. Differentiating α in Equation 1 and $m = p(1 - \ln p)$ gives

$$\alpha_p = \frac{\alpha}{p(1-p)}, \quad \alpha_{\mu_0} = -\frac{p}{(1-p)\mu_0^2}, \quad \frac{dm}{dp} = -\ln p. \quad (26)$$

The derivatives of platform-access probability enter the mean constraint in Equation 24. For its derivative with respect to μ_0 , the integral in Equation 3 contributes

$$\frac{\partial}{\partial \mu_0} \int_{\mu_0}^{\mu'} t dF(t) = -\mu_0 f(\mu_0). \quad (27)$$

Substituting Equation 26 and Equation 27 into the derivatives of Equation 3 yields

$$\frac{\partial}{\partial p} \mathbf{E}[\bar{q}] = \frac{\alpha}{p(1-p)} \int_{\mu_0}^{\mu'} t dF(t), \quad (28)$$

$$\begin{aligned} \frac{\partial}{\partial \mu_0} \mathbf{E}[\bar{q}] &= -f(\mu_0) + \alpha_{\mu_0} \int_{\mu_0}^{\mu'} t dF(t) + (1-\alpha)\mu_0 f(\mu_0) \\ &= -\frac{1-\mu_0}{1-p} f(\mu_0) - \frac{p}{(1-p)\mu_0^2} \int_{\mu_0}^{\mu'} t dF(t). \end{aligned} \quad (29)$$

The revenue derivatives follow from Equation 5:

$$\Pi_p = -\frac{\mu' [1 - F(\mu')]}{\mu_0}, \quad \Pi_{\mu_0} = \frac{p\mu' [1 - F(\mu')]}{\mu_0^2} > 0. \quad (30)$$

Using $dm/dp = -\ln p$ from Equation 26, the two first-order conditions are

$$0 = \Pi_p - \lambda \left[\ln p \mathbf{E}[\bar{q}] + (1-m) \frac{\partial}{\partial p} \mathbf{E}[\bar{q}] \right], \quad (31)$$

$$0 = \Pi_{\mu_0} - \lambda \left[(1-m) \frac{\partial}{\partial \mu_0} \mathbf{E}[\bar{q}] + (1-\mu_0) f(\mu_0) \right]. \quad (32)$$

Since $\Pi_{\mu_0} > 0$ by Equation 30, Equation 32 implies $\lambda \neq 0$.

Adding p times Equation 31 to μ_0 times Equation 32 gives

$$\begin{aligned} 0 &= p\Pi_p + \mu_0\Pi_{\mu_0} \\ &\quad - \lambda \left\{ p \ln p \mathbf{E}[\bar{q}] + (1-m) \left[p \frac{\partial}{\partial p} \mathbf{E}[\bar{q}] + \mu_0 \frac{\partial}{\partial \mu_0} \mathbf{E}[\bar{q}] \right] \right. \\ &\quad \left. + \mu_0(1-\mu_0)f(\mu_0) \right\}. \end{aligned} \quad (33)$$

The revenue terms in Equation 33 cancel by Equation 30. Substituting Equation 28 and Equation 29 into Equation 33, dividing by $-\lambda \neq 0$, and multiplying by $(1 - p)/p$ yields

$$0 = (1 - p) \ln p \mathbf{E}[\bar{q}] - (1 - m)(1 - \alpha) \int_{\mu_0}^{\mu'} t dF(t) - \ln p \mu_0(1 - \mu_0)f(\mu_0). \quad (34)$$

In Equation 34, replace $(1 - \alpha) \int_{\mu_0}^{\mu'} t dF(t)$ by $1 - F(\mu_0) - \mathbf{E}[\bar{q}]$, using Equation 3. We obtain

$$0 = (1 - p) \ln p \mathbf{E}[\bar{q}] - (1 - m)[1 - F(\mu_0) - \mathbf{E}[\bar{q}]] - \ln p \mu_0(1 - \mu_0)f(\mu_0). \quad (35)$$

Finally substitute $\mathbf{E}[\bar{q}] = \frac{1}{1-m} \int_{\mu_0}^1 (1 - t) dF(t)$ from Equation 4 into Equation 35. With $m = p(1 - \ln p)$, this eliminates the remaining dependence on μ' :

$$\begin{aligned} & (-\ln p - 1 + p) \int_{\mu_0}^1 (1 - t) dF(t) + (1 - m)^2[1 - F(\mu_0)] \\ & = -(1 - m) \ln p \mu_0(1 - \mu_0)f(\mu_0). \end{aligned} \quad (36)$$

A bound uniform over the remaining prices. An optimal first-form mechanism must have $3/10 < p < 1/2$. We now use Equation 36 to bound its lower cutoff by $3p/(1 + 2p)$. First we establish the strict inequality for the uniform distribution, including the endpoints $3/10$ and $1/2$. Continuity on this closed interval then gives a positive minimum difference between the proposed bound and the uniform cutoff. The density estimates below will preserve that difference for all remaining prices with a single sufficiently small ε . For the uniform distribution, Equation 36 gives

$$\mu_0 = \frac{-\ln p - 1 + p + 2(1 - m)^2}{-\ln p - 1 + p - 2(1 - m) \ln p} < \frac{3p}{1 + 2p}. \quad (37)$$

The strict scalar inequality is proved in Section 2. If $\|f - 1\|_\infty < \varepsilon < 1$, then $1 - \varepsilon < f(t) < 1 + \varepsilon$ for every t . Integrating the upper bound gives the first two inequalities below; evaluating the lower bound at μ_0 gives the third:

$$\int_{\mu_0}^1 (1 - t) dF(t) \leq \frac{1 + \varepsilon}{2} (1 - \mu_0)^2, \quad (38)$$

$$1 - F(\mu_0) \leq (1 + \varepsilon)(1 - \mu_0), \quad (39)$$

$$f(\mu_0) \geq 1 - \varepsilon. \quad (40)$$

Both coefficients on the left of Equation 36 are positive, as is the coefficient of $f(\mu_0)$ on the right. Substituting Equation 38, Equation 39, and Equation 40 into Equation 36 therefore gives

$$\begin{aligned} & -2(1-\varepsilon)(1-m)\ln p \mu_0(1-\mu_0) \\ & \leq (1+\varepsilon) \left[(-\ln p - 1 + p)(1-\mu_0)^2 + 2(1-m)^2(1-\mu_0) \right]. \end{aligned} \quad (41)$$

Divide Equation 41 by $1-\mu_0 > 0$, then move the terms involving μ_0 to the left:

$$\begin{aligned} & \mu_0 \left[(1+\varepsilon)(-\ln p - 1 + p) - 2(1-\varepsilon)(1-m)\ln p \right] \\ & \leq (1+\varepsilon) \left[-\ln p - 1 + p + 2(1-m)^2 \right]. \end{aligned} \quad (42)$$

The coefficient of μ_0 in Equation 42 is positive. Dividing by it yields

$$\mu_0 \leq \frac{(1+\varepsilon)[-\ln p - 1 + p + 2(1-m)^2]}{(1+\varepsilon)(-\ln p - 1 + p) - 2(1-\varepsilon)(1-m)\ln p}. \quad (43)$$

The difference between $3p/(1+2p)$ and the fraction in Equation 37 is continuous and strictly positive on $[3/10, 1/2]$, so its minimum is positive. The right side of Equation 43 converges uniformly to that fraction as ε tends to zero, because its denominator stays bounded away from zero on this interval. We can therefore choose one $0 < \varepsilon \leq 1/100$ satisfying Equation 11 and small enough that $\mu_0 < 3p/(1+2p)$ for every remaining price. Finally, Equation 1 gives

$$\alpha > \frac{1}{3} \iff 3p(1-\mu_0) > (1-p)\mu_0 \iff \mu_0 < \frac{3p}{1+2p}. \quad (44)$$

This proves ?? of the main paper. □

The calculation for a uniform prior is given in Section 3.

2 Scalar Inequality for ??

For the uniform distribution, Section 1 requires $\mu_0 < 3p/(1+2p)$ for every $p \in [3/10, 1/2]$. Substituting the fraction in Equation 37 and clearing positive denominators gives the equivalent inequality

$$\begin{aligned} & 3p[-\ln p - 1 + p - 2(1-m)\ln p] \\ & - (1+2p)[- \ln p - 1 + p + 2(1-m)^2] > 0. \end{aligned} \quad (45)$$

Substitute $m = p(1 - \ln p)$ into Equation 45. At each fixed price, its left side is a quadratic in the logarithm $-\ln p$, with leading coefficient $-4p^2(p + 2) < 0$. We compare this concave quadratic at the two bounds

$$\frac{2(1-p)}{1+p} \leq -\ln p \leq \frac{1-p}{\sqrt{p}}, \quad 0 < p < 1. \quad (46)$$

The lower bound follows by integrating the convex function $1/x$ on $[p, 1]$. For the upper bound, set $x = 1/\sqrt{p} \geq 1$: the difference $x - 1/x - 2 \ln x$ vanishes at one and has derivative $(x - 1)^2/x^2 \geq 0$.

After eliminating m , keep p fixed and substitute each bound in Equation 46 for $-\ln p$. The quadratic lies above the chord joining these two values, so positivity at both proves Equation 45.

Substituting the lower bound $2(1-p)/(1+p)$ gives

$$\frac{(1-p)^3}{(1+p)^2} (4p^2 + 9p - 3) > 0. \quad (47)$$

The prefactor is positive, and the bracket is positive at $p = 3/10$ and increasing for $p \geq 3/10$.

Substituting the upper bound $(1-p)/\sqrt{p}$ gives

$$\frac{(1-\sqrt{p})^3(1+\sqrt{p})^2}{\sqrt{p}} [4p^2 - 4p^{3/2} + 8p - 2\sqrt{p} - 1] > 0. \quad (48)$$

Here too the prefactor is positive. The bracket is increasing in $\sqrt{p} \geq 1/2$: its first derivative at $1/2$ is 5 and its second derivative is $48(\sqrt{p} - 1/4)^2 + 13 > 0$. Its value at $p = 3/10$ is $44/25 - (16/5)\sqrt{3/10} > 0$, since $\sqrt{3/10} < 11/20$. Thus Equation 45 holds on $[3/10, 1/2]$, establishing Equation 37.

3 Uniform Prior– ??

We now derive the revenue-maximizing mechanism within the two single-price forms in ?? for a uniform prior. Let $F(\mu) = \mu$. By the mean argument in ??, it suffices within these forms to consider ?? with

$$m = \mathbf{E}H_0^p = p(1 - \ln p). \quad (49)$$

First, we look at the second form in ??:

$$(q_1(\mu), q_0(\mu)) = \begin{cases} (0, 0), & \mu < \mu_0, \\ (1, 0), & \mu \in [\mu_0, \mu'), \\ (1, 1), & \mu \geq \mu'. \end{cases}$$

The mean constraint binding with no point mass at p gives the following problem at a fixed price:

$$\begin{aligned} \max_{\mu_0, \mu'} \quad & (1-p) \int_{\mu_0}^1 (2\mu - 1) d\mu - 2p \int_{\mu'}^1 (1 - \mu) d\mu \\ \text{s.t.} \quad & 0 \leq \mu_0 \leq \mu' \leq 1, \quad \mu' \geq p, \\ & \int_{\mu_0}^1 \mu d\mu = \frac{m}{1-m} \int_{\mu'}^1 (1 - \mu) d\mu. \end{aligned} \tag{50}$$

The last inequality on the cutoffs is platform full participation in ?? of the main paper. Using the mean equality in Equation 50, the objective becomes

$$\begin{aligned} & \int_{\mu_0}^1 \left[2\mu \left(1 - \frac{p}{m} \right) - (1-p) \right] d\mu \\ & = (1-p)\mu_0(1-\mu_0) - \frac{p(1-m)}{m}(1-\mu_0^2). \end{aligned} \tag{51}$$

This expression is strictly concave in μ_0 , with unconstrained maximizer

$$\mu^*(p) = \frac{1-p}{2(1-p/m)}. \tag{52}$$

The mean equality determines the other cutoff:

$$\mu' = 1 - \sqrt{\frac{1-m}{m}(1-\mu_0^2)}. \tag{53}$$

In particular, $\mu_0 \leq \mu'$ requires $\mu_0 \leq 2m - 1$. If the cutoff in Equation 52 satisfies the constraints in Equation 50, it is optimal at that price. If instead it exceeds $2m - 1$, the optimal feasible cutoffs coincide. Substituting $\mu_0 = \mu'$ into Equation 53 gives the unique solution below one,

$$\mu_0 = \mu' = 2m - 1.$$

At these prices, Equation 51 reduces to

$$R(p) = 2(1 - m)[2m - (1 + p)]. \quad (54)$$

The first-order condition for the price is transcendental, so we calculate the optimum numerically. Maximizing Equation 51 over the feasible cutoffs and prices gives revenue approximately 0.0646058, attained at

$$p \simeq 0.436431, \quad \mu_0 = \mu' \simeq 0.596573. \quad (55)$$

Next, consider the first form in ??:

$$(q_1(\mu), q_0(\mu)) = \begin{cases} (0, 0), & \mu < \mu_0, \\ \left(\frac{p(1 - \mu_0)}{(1 - p)\mu_0}, 1 \right), & \mu \in [\mu_0, \mu'], \\ (1, 1), & \mu \geq \mu'. \end{cases}$$

Write $\alpha = p(1 - \mu_0)/[(1 - p)\mu_0]$, as in Equation 1. We solve

$$\begin{aligned} \max_{\mu_0, \mu'} \quad & (1 - p)\alpha \int_{\mu_0}^{\mu'} (2\mu - 1) d\mu + (1 - p) \int_{\mu'}^1 (2\mu - 1) d\mu \\ & - 2p \int_{\mu_0}^1 (1 - \mu) d\mu \\ \text{s.t.} \quad & p \leq \mu_0 \leq \mu' \leq 1, \\ & \alpha \int_{\mu_0}^{\mu'} \mu d\mu + \int_{\mu'}^1 \mu d\mu = \frac{m}{1 - m} \int_{\mu_0}^1 (1 - \mu) d\mu. \end{aligned} \quad (56)$$

Using its mean equality, the objective in Equation 56 can be rewritten as

$$\begin{aligned} & \alpha \int_{\mu_0}^{\mu'} \left[2\mu \left(1 - \frac{p}{m} \right) - (1 - p) \right] d\mu \\ & + \int_{\mu'}^1 \left[2\mu \left(1 - \frac{p}{m} \right) - (1 - p) \right] d\mu. \end{aligned} \quad (57)$$

For $\mu_0 > p$, the mean equality in Equation 56 determines

$$\mu' = h(\mu_0) := \sqrt{\mu_0^2 + \frac{(1 - p)\mu_0(1 - \mu_0)}{(\mu_0 - p)(1 - m)}} (1 + \mu_0 - 2m). \quad (58)$$

Here the dependence of h on the price is through p and Equation 49. Feasibility requires $\mu_0 \geq 2m - 1$ and $h(\mu_0) \leq 1$, in addition to $\mu_0 > p$. Substituting Equation 58 into Equation 57, or equivalently into Equation 5, leaves the two-variable problem

$$\begin{aligned} \max_{p, \mu_0} \quad & \left(1 - \frac{p}{\mu_0}\right) h(\mu_0)[1 - h(\mu_0)] \\ \text{s.t.} \quad & 0 < p < \mu_0 < 1, \quad \mu_0 \geq 2m - 1, \quad h(\mu_0) \leq 1. \end{aligned} \quad (59)$$

At $\mu_0 = p$, revenue is zero. Equal cutoffs are included in the preceding form. Numerical maximization gives **revenue approximately** 0.0652205, attained at

$$p^* \simeq 0.410377, \quad \mu_0 \simeq 0.573355, \quad \mu' \simeq 0.643369. \quad (60)$$

Thus the revenue-maximizing allocation within the two forms is, to five decimal places,

$$(q_1(\mu), q_0(\mu)) \simeq \begin{cases} (0, 0), & \mu < 0.57335, \\ (0.51791, 1), & \mu \in [0.57335, 0.64337], \\ (1, 1), & \mu \geq 0.64337. \end{cases} \quad (61)$$

The equations above determine the unrounded parameters. To describe their induced belief law, let

$$\bar{v} = \frac{\alpha \mu'}{\alpha \mu' + 1 - \mu'} \simeq 0.483021, \quad w = \int_{\mu_0}^{\mu'} [\alpha t + (1 - t)] dt + 1 - \mu'. \quad (62)$$

The CDF of beliefs induced by platform access is

$$G_q(z) = \begin{cases} 0, & z < p^*, \\ \frac{1}{w} \int_{\mu_0}^{\frac{z}{z + \alpha(1-z)}} [\alpha t + (1 - t)] dt, & z \in [p^*, \bar{v}), \\ \frac{1}{w} \int_{\mu_0}^{\mu'} [\alpha t + (1 - t)] dt, & z \in [\bar{v}, \mu'), \\ \frac{1}{w} \left[\int_{\mu_0}^{\mu'} [\alpha t + (1 - t)] dt + z - \mu' \right], & z \in [\mu', 1]. \end{cases} \quad (63)$$

By Equation 56, this distribution has the same mean as $H_0^{p^*}$. Since $\alpha > 1/3$, the integrated-CDF argument in ?? of the main appendix gives $H_0^{p^*} \succeq_{\text{mps}} G_q$. By ??, additional private information implements this allocation at the stated price without changing its fees or revenue.

4 A Two-Price Platform with Uniform Types

Let $F(\mu) = \mu$. We construct a mechanism with two nontrivial prices whose revenue exceeds that obtained within the two single-price forms in [Section 3](#). Purchases at the recommended prices are enforceable, as in ?? of the main paper.

Allocation and prices. Let $q_\theta^j(\mu)$ be the probability that type μ with valuation θ receives price recommendation p_j , for $j = 1, 2$. The remaining probability is exclusion. Set $\mu_0 = 11/20$, $\mu_1 = 3/5$, and $\mu' = 3/4$, and choose

Types	(q_1^1, q_0^1)	(q_1^2, q_0^2)
$[0, \mu_0)$	$(0, 0)$	$(0, 0)$
$[\mu_0, \mu_1)$	$(0, 0)$	$(1, \frac{1}{4})$
$[\mu_1, \mu')$	$(\frac{9}{20}, 1)$	$(\frac{11}{20}, 0)$
$[\mu', 1]$	$(1, 1)$	$(0, 0)$.

(64)

Thus buyers in $[\mu_1, \mu')$ always obtain access, but their price recommendation depends on their valuation. Write $\bar{q}^j(\mu) = \mu q_1^j(\mu) + (1 - \mu)q_0^j(\mu)$. Integrating [Equation 64](#) gives the access probabilities and conditional means

$$\mathbf{E}[\bar{q}^1] = \frac{5509}{16000}, \quad m_1 = \frac{4229}{5509}, \quad \mathbf{E}[\bar{q}^2] = \frac{359}{4000}, \quad m_2 = \frac{1351}{1436}. \quad (65)$$

Choose the unique prices satisfying $p_j(1 - \ln p_j) = m_j$. They obey

$$0.4012 < p_1 < 0.4013, \quad 0.6762 < p_2 < 0.6763. \quad (66)$$

These bounds follow by substituting their endpoints into the strictly increasing function $p(1 - \ln p)$.

Incentives and participation. The slope of gross deviation utility, summing across recommendations, is

$$K(\mu) = \sum_{j=1}^2 [(1 - p_j)q_1^j(\mu) + p_j q_0^j(\mu)] = \begin{cases} 0, & \mu < \mu_0, \\ 1 - \frac{3}{4}p_2, & \mu_0 \leq \mu < \mu_1, \\ 1 - \frac{11}{20}(p_2 - p_1), & \mu_1 \leq \mu < \mu', \\ 1, & \mu \geq \mu'. \end{cases} \quad (67)$$

By [Equation 66](#), these slopes increase from zero to one. Set

$$\phi(\mu) = \sum_{j=1}^2 [(1 - p_j)\mu q_1^j(\mu) - p_j(1 - \mu)q_0^j(\mu)] - \int_0^\mu K(t) dt. \quad (68)$$

The envelope argument in ?? now gives incentive compatibility and nonnegative truthful utility for every type. Before additional information, participating buyers' beliefs are at least $27/67 > p_1$ in the first segment and $44/53 > p_2$ in the second. We next verify that each segment can be supplied with private information inducing the unit-elastic distribution at its recommended price; its support then ensures full participation.

Seller obedience. For the first segment, put $p = p_1$ and $\alpha = 9/20$. Here $q_0^1 = 1$ for every type above μ_1 , while $q_1^1 = 9/20$ on $[\mu_1, \mu']$ and one above μ' . The pointwise bound in ?? of the main paper gives an upper bound on its integrated CDF. Define the difference between the unit-elastic integrated CDF and this upper bound by

$$D(z) = \mathbf{E}[\bar{q}^1][z - m_1 - p \ln z] - \int_{\mu_1}^1 [z(1-t) - (1-z)\alpha t]_+ dt. \quad (69)$$

Here $D(p) = D'(p) = D(1) = 0$: the integral vanishes below $z = 27/67$, and at one its value is $(1 - \mu_1)^2/2 = (1 - m_1)\mathbf{E}[\bar{q}^1]$. Below $27/67$, $D'' > 0$; above it, ?? applies with density one. Since $\alpha > 1/3$, ?? shows that D'' can change sign only from positive to negative. Thus D first lies above its zero tangent at p and then above the chord to $D(1) = 0$. Hence $D \geq 0$, establishing convex order for the first segment.

For the second segment, write $p = p_2$ and let G be its conditional belief law. Its continuous part is supported on $[44/53, 6/7]$, with

$$G'(z) = \frac{(1/4)^2}{\mathbf{E}[\bar{q}^2][1 - (3/4)z]^3}. \quad (70)$$

The remaining mass, $891/1436$, is at one. The ratio $z^2 G'(z)/p$ increases strictly on the continuous part. Consequently, $G - H_0^p$ can cross zero there only from below. It remains nonnegative after $6/7$, since its value at $1-$ is $p - 891/1436 > 0$. Together with equal means, this single crossing proves convex order. Applying ?? separately to the two segments implements both prices without changing allocations, fees, or reporting incentives.

Revenue comparison. Integrating Equation 68 yields

$$\Pi = \frac{99}{400} - \frac{2969}{8000}p_1 - \frac{47}{1000}p_2 > \frac{1068503}{16000000} > \frac{1}{15}, \quad (71)$$

The first inequality follows from Equation 66; revenue is approximately 0.06680.

This exceeds the revenue approximately 0.0652205 obtained within the two single-price forms in Equation 60.